Tutorial - User Guide

This tutorial shows you how to use **FastAPI** with most of its features, step by step.

Each section gradually builds on the previous ones, but it's structured to separate topics, so that you can go directly to any specific one to solve your specific API needs.

It is also built to work as a future reference so you can come back and see exactly what you need.

Run the code

All the code blocks can be copied and used directly (they are actually tested Python files).

To run any of the examples, copy the code to a file `main.py`, and start `fastapi dev`:

```bash
$ fastapi dev

  FastAPI   Starting development server 🚀

            Searching for package file structure
from directories
            with __init__.py files
            Importing from
/home/user/code/awesomeapp

   module   🐍 main.py

     code   Importing the FastAPI app object from
the module with
            the following code:

            from main import app

      app   Using import string: main:app

   server   Server started at http://127.0.0.1:8000
```

Ask AI

```
    server   Documentation at
http://127.0.0.1:8000/docs

       tip   Running in development mode, for
production use:
             fastapi run

             Logs:

      INFO   Will watch for changes in these
directories:
             ['/home/user/code/awesomeapp']
      INFO   Uvicorn running on
http://127.0.0.1:8000 (Press CTRL+C
             to quit)
      INFO   Started reloader process [383138] using
WatchFiles
      INFO   Started server process [383153]
      INFO   Waiting for application startup.
      INFO   Application startup complete.

                                          restart ↻
```

It is **HIGHLY encouraged** that you write or copy the code, edit it and run it locally.

Using it in your editor is what really shows you the benefits of FastAPI, seeing how little code you have to write, all the type checks, autocompletion, etc.

Install FastAPI

The first step is to install FastAPI.

Make sure you create a virtual environment, activate it, and then **install FastAPI**:

```bash
```

Note

When you install with `pip install "fastapi[standard]"` it comes with some default optional standard dependencies, including `fastapi-cloud-cli`, which allows you to deploy to FastAPI Cloud.

If you don't want to have those optional dependencies, you can instead install `pip install fastapi`.

If you want to install the standard dependencies but without the `fastapi-cloud-cli`, you can install with `pip install "fastapi[standard-no-fastapi-cloud-cli]"`.

Tip

FastAPI has an official extension for VS Code (and Cursor), which provides a lot of features, including a path operation explorer, path operation search, CodeLens navigation in tests (jump to definition from tests), and FastAPI Cloud deployment and logs, all from your editor.

Advanced User Guide

There is also an **Advanced User Guide** that you can read later after this **Tutorial - User guide**.

The **Advanced User Guide** builds on this one, uses the same concepts, and teaches you some extra features.

But you should first read the **Tutorial - User Guide** (what you are reading right now).

It's designed so that you can build a complete application with just the **Tutorial - User Guide**, and then extend it in different ways, depending on your needs, using some of the additional ideas from the **Advanced User Guide**.